

SAIKAT DAS

Backend-Focused Development | DevOps | AI/ML Engineering

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Portfolio | GitHub



Professional Summary

Computer Science and Engineering graduate with a strong interest in backend development, DevOps, and AI/ML engineering. Experienced in developing and deploying web applications using Node.js, Express.js, Django, REST APIs, PostgreSQL, MongoDB, MySQL, Redis, and cloud deployment platforms. Familiar with backend architecture, authentication, caching, database design, deployment workflows, networking fundamentals, and basic DevOps practices. Knowledgeable in machine learning, deep learning, computer vision, vision-language models, LLM concepts, prompt engineering, fine-tuning concepts, and Responsible AI.

Technical Skills

Backend-Focused	DevOps	AI/ML Engineering
• Node.js and Express.js • TypeScript and Nest.js • REST API development • PostgreSQL, MongoDB and MySQL • Redis caching • JWT authentication • Email Authentication • Database design and transactions • Load-balancing concepts • System design fundamentals • Scalable backend concepts • API security and optimization	• Git and GitHub • Linux fundamentals • Docker fundamentals • CI/CD fundamentals • AWS fundamentals • Vercel and Render • Firebase and Cloudinary • Environment configuration • Application deployment • Logging and monitoring concepts • Basic server management • Deployment troubleshooting	• Machine Learning • Deep Learning • Computer Vision • Vision-Language Models • Natural Language Processing • Large Language Models • Fine-tuning concepts • Prompt engineering • Responsible AI • Model evaluation • Data preprocessing • PyTorch and TensorFlow
Programming Languages	JavaScript, typescript, Python, C++, SQL, MATLAB, Java, and PHP (basic)	
Computer Science Fundamentals	DSA, Object-Oriented Programming, Operating Systems, Computer Networks, Computer Architecture, Database Management Systems, and Software Engineering	
Development Tools	Visual Studio Code, Jupyter Notebook, Postman, GitHub, XAMPP, and database management tools	
Problem Solving	Solved 200+ programming problems on Codeforces, LeetCode, and other online problem-solving platforms	

Education

B.Sc. in Computer Science and Engineering	Graduated 2026
Ahsanullah University of Science and Technology (AUST), Dhaka	CGPA: 3.32/4.00
CGPA in Last Four Semesters: 3.65/4.00	
Higher Secondary Certificate (HSC) , Chittagong College	2020 GPA: 5.00/5.00
Secondary School Certificate (SSC) , Saint Placid's High School	2018 GPA: 5.00/5.00

Development Projects

CommerceX – AI-Powered Multi-Vendor Marketplace

GitHub | Live Website

Stack: Node.js, PostgreSQL, Redis, Socket.IO, React, Vite, Docker and Cloudinary

Why this project: Demonstrates complete e-commerce workflows, REST API design, inventory handling, and real-time communication.

- Built customer, seller, and administrator workflows with JWT authentication, role-based access, product variants, checkout, order fulfilment, reviews, and shipment tracking.
- Used transactional stock updates and Redis caching for reliable inventory, with Socket.IO for real-time admin-seller messaging.

Dhaka Tesla Pool – Shared Ride Platform

GitHub | Live Website

Stack: React.js, Vite, Node.js, Express.js, MongoDB Atlas, Socket.IO, JWT, Cloudinary and Docker

Why this project: Demonstrates route-matching algorithms, seat- and distance-based fare calculation, and real-time ride updates.

- Built a ride-sharing system that groups passengers travelling along compatible routes and calculates individual fares from shared distance and booked seats.
- Used Socket.IO for live communication and REST APIs for passenger, ride, authentication, and media workflows.

Stremars – YouTube-Style Video Sharing Platform

GitHub | Live Website

Stack: React.js, Vite, Node.js, Express.js, MongoDB, Mongoose, Cloudinary, Multer and JWT

Why this project: Demonstrates large media-file handling, secure uploads, and the data flow behind a video-sharing platform.

- Implemented video upload, viewing, search, comments, likes, subscriptions, watch history, creator profiles, and community posts.
- Used Multer and Cloudinary to manage videos, thumbnails, avatars, and cover images while storing metadata in MongoDB.
- Built user authentication, room management, reservations, payments, inquiries, and administrative features.
- Designed a relational MySQL database with foreign-key constraints and transaction-based booking operations.

AI/ML Engineering Project

Edge-AI Waste Classification – Real-Time Smart Bin

GitHub | Demo Video

Stack: Python, TensorFlow Lite, Raspberry Pi, OpenCV, Firebase and Flask

- Developed and evaluated a real-time Edge-AI waste-classification system using computer vision models deployed on Raspberry Pi.
- Integrated image capture, biometric authentication, fill-level monitoring, hardware control, secure logging, and Firebase-based remote monitoring.

Publication Under Review

Saikat Das, et al. “A Deployment and Resource Aware Edge AI Framework for Real-Time Waste Classification with Biometric Authentication and Secure Logging” – under review, submitted April 2026.

Training & Certifications

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- **OpenCV University:** Python, PyTorch, TensorFlow-Keras, and Vision-Language Model Bootcamp 2025–2026
 - **Google Cloud Skills Boost:** Prompt Design, Responsible AI, Introduction to LLMs, Introduction to Generative AI, and ML Engineer Guide 2025–2026
 - **MathWorks:** MATLAB Onramp, Machine Learning Onramp, and Deep Learning Onramp 2026
 - **IBM/Coursera:** Introduction to Data Engineering February 2026

Additional Information

Languages	Bengali – Native; English – Intermediate (B1)
Strengths	Backend-focused problem-solving, quick learning, teamwork, technical documentation, adaptability, and attention to detail
Target Roles	Backend Developer, Node.js Developer, Software Engineer, Junior DevOps Engineer, and AI/ML Engineer

References

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Signature of Saikat Das